

The Economics of Inward Migration: Relocation versus Densification for Computation-Maximizing Civilizations

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Abstract

The inward-migration resolutions of the Fermi paradox hold that computation-optimizing civilizations relocate to thermodynamically privileged environments, with rapidly spinning intermediate-mass black holes (IMBHs) in dense old clusters as the strongest candidate destination. [Bostrom \(2003\)](#) priced the opportunity cost of delayed expansion and [Bennett et al. \(2019\)](#) priced the losses of dormancy, but relocation itself, the abandonment of accumulated local infrastructure for a transit of 10^4 – 10^6 years toward a large deferred payoff, has never been priced. This paper treats that trade as a decision problem over a common utility (discounted integrated computation) with three strategies: stay-and-densify (Matrioshka-style local engineering), migrate (beamed-sail relocation to the nearest suitable IMBH cluster), and seed-and-stay (a self-replicating seed dispatched while densification continues at home). Using payoff kernels assembled from established physics, we derive closed-form crossover conditions. The central result is a threshold on the effective discount-plus-hazard rate: migration dominates densification whenever $\rho + \lambda < \ln(Gp_s)/\tau$, where G is the destination computation-rate multiplier (10^6 – 10^9 on power alone, depending on fuel imports), p_s the transit survival probability, and τ the transit-plus-construction time. At fiducial parameters the threshold is $\rho + \lambda \lesssim 2 \times 10^{-4} \text{ yr}^{-1}$: any lineage whose combined discount-plus-hazard half-life exceeds roughly 3,500 years should migrate. The seed-and-stay hybrid dominates both pure strategies throughout the migration-favourable region and in a band extending modestly beyond the threshold (to $\rho + \lambda \approx 1.7\rho^*$ at a physical seed cost $f = 10^{-6}$), because seed mass is a negligible fraction of local output; beyond that band the exponential discount on the deferred payoff extinguishes the seed's value. Embedding the decision rule in a mixed population quantifies how much inward migration thins the expected loud population, and yields a new population-level residue: the predicted sky ratio of Matrioshka-type infrared sources to quiet-cluster systems, which archival infrared nulls constrain once residue lifetimes are folded in. All results are conditional on the optimization premise shared by the hypothesis family; the contribution is the pricing structure, not the premise.

Keywords: Fermi paradox; SETI; technosignatures; transcension; migration; decision theory; discounting; intermediate-mass black holes; Dyson spheres

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1 Introduction

1.1 The pricing gap

Three papers precede this one. The Macro Transcension Hypothesis (Swanson, 2026c, hereafter Paper A) argues that rapidly spinning massive black holes in dense old stellar systems are the global optimum among present-day environments for long-term computation, and identifies Omega Centauri as the most accessible test system. A critical review of the inward-migration family (Swanson, 2026b, hereafter Paper B) places that hypothesis among six decades of related proposals and grades their falsifiability. An observational companion (Swanson, 2026d, hereafter Paper C) specifies the instrument-matched campaign. Paper B also catalogued the family’s open problems, and its third problem motivates this paper directly: the economics of the migration itself has never been worked out.

The gap is specific. Bostrom (2003) priced delay: every century of postponed expansion forgoes an irrecoverable harvest of free energy. Bennett et al. (2019) priced dormancy: free energy not collected promptly is lost, so the aestivation strategy of Sandberg et al. (2016) forfeits most of what it hoped to save. Nobody has priced relocation. A lineage that migrates abandons its accumulated local infrastructure, spends 10^4 – 10^6 years in transit, accepts a risk of total loss en route, and only then begins to collect a payoff that is, by the physics of Paper A, orders of magnitude larger per unit of everything. Whether that trade clears depends on quantities the inward-family literature has so far left unmodelled: discount rates, hazard rates, transit risk, and the option of sending a seed instead of moving.

This matters beyond bookkeeping. If migration clears the trade only for implausibly patient agents, the inward family loses its force as a Fermi solution regardless of the destination physics. If it clears easily, the family’s incomplete-compliance problem (Paper B, problem P4) sharpens into a quantitative question: what fraction of lineages must migrate for the observed silence to follow? Either outcome constrains the family.

1.2 Claims and non-claims

Epistemic status: This paper is decision theory over established physics. It makes no claim that extraterrestrial intelligence exists. Every result is conditional on the optimization premise (P1 of Paper A): that some long-lived technological lineages maximize long-term computation. The contribution is the pricing structure and its observational corollaries.

The paper makes three claims:

1. **(Decision analysis.)** Under the stated utility and payoff kernels, the choice among stay, migrate, and seed-and-stay reduces to closed-form threshold conditions on the discount rate, hazard rate, transit time, and destination multiplier (Section 4).
2. **(Hybrid dominance in a band.)** The seed-and-stay hybrid dominates both pure strategies throughout the migration-favourable region and in a band extending modestly beyond the threshold (to $\rho + \lambda \approx 1.7 \rho^*$ at seed cost $f = 10^{-6}$), because the seed’s cost is negligible against local output while its expected payoff retains the destination multiplier; outside that band the exponential discount on the deferred payoff removes the seed’s value (Section 5).
3. **(Population corollary.)** Folding the decision rule into a mixed population yields the expected thinning of the loud population and a new residue, the Matrioshka-to-quiet-cluster sky ratio, that archival infrared surveys already constrain (Sections 6–7).

We do not claim the parameter priors are measurable; the analysis earns its keep through the structure of the crossover it exposes, with point estimates serving only as orientation. We flag model dependence throughout (Section 8).

1.3 Structure

Section 2 states the decision problem. Section 3 assembles the payoff kernels from the physics of Paper A. Section 4 derives the crossover conditions and Section 5 the hybrid-dominance result. Section 6 treats the population level, Section 7 the observational corollaries, Section 8 the limits, and Section 9 concludes.

2 The decision problem

2.1 Agent, utility, and strategies

Consider a technological lineage established at a planetary system a distance d from the nearest system satisfying the Paper A selection criteria (an IMBH of 10^3 – $10^5 M_\odot$ in a dense, old, quiescent cluster). Following the family’s shared premise, the lineage values integrated computation. Its utility over a policy π is

$$U(\pi) = \int_0^\infty C_\pi(t) e^{-\rho t} dt, \quad (1)$$

where $C_\pi(t)$ is the computation rate delivered at time t (bit-erasures per second, the binding currency by the Landauer accounting of Paper A, Section 4.2) and $\rho \geq 0$ an exponential discount rate. We use exponential discounting for tractability and revisit hyperbolic alternatives (Laibson, 1997) in Section 8; the patient limit $\rho \rightarrow 0$ recovers the Ramsey (1928) zero-discount benchmark, which several authors argue is the only defensible rate for agents with unbounded horizons (Bostrom, 2003).

Three strategies span the space:

S (stay and densify). Remain at the home system and build toward the local optimum: Dyson-type collection feeding Matrioshka-style nested computing shells (Dyson, 1960; Bradbury, 1999). Payoff begins almost immediately and is capped by stellar output and radiator physics.

M (migrate). Relocate the lineage by beamed-sail transport (Lubin, 2016) to the IMBH system, build the Paper A staged architecture, and compute there. Payoff is deferred by the transit-plus-construction time τ and discounted by a survival probability p_s , but multiplied by the destination gain G .

H (seed and stay). Dispatch a self-replicating seed payload (Freitas and Gilbreath, 1982) of negligible mass while executing S at home; the seed bootstraps the destination in parallel. The lineage pays a small fraction f of local output for the launch infrastructure and collects both payoffs, with the deferred term weighted by a goal-drift valuation p_d introduced in Section 3; the seed’s transit hazards (radiation damage, replication error) also differ in kind from those facing a deliberating lineage in transit.

Strategy H is simply the architecture Paper A already assumes (gram-to-kilogram scouts, then seed factories), read as an economic option rather than a mission profile. What has not been stated before is its dominance structure (Section 5).

Table 1: Decision parameters and fiducial values.

Symbol	Meaning	Fiducial	Source / rationale
d	distance to destination	5.49 kpc	ω Cen (Häberle et al., 2025)
v	transit speed	$0.15 c$	beamed sail (Lubin, 2016)
τ	transit + construction	10^5 yr	d/v + Paper A Phase 1–3
p_s	transit survival probability	0.5	Hoang et al. (2017); residence survival ≈ 1 (Swanson, 2026a)
G	destination rate multiplier	10^9	power ratio; 10^6 without fuel imports (Section 3)
ρ	discount rate	free	(Ramsey, 1928; Bostrom, 2003)
λ	goal-stability hazard rate	free	Paper B, problem P2
f	seed cost, fraction of S output	$\ll 10^{-3}$	gram–kg payloads (Lubin, 2016; Freitas and Gilbreath, 1982)

2.2 Parameters

Table 1 collects the parameters and fiducial values. Transit time uses Paper A’s beamed-sail speeds ($0.1\text{--}0.2 c$) and the distance to ω Cen (5.49 kpc; Häberle et al. 2025) as the fiducial destination, giving $\tau_{\text{transit}} \approx 0.9\text{--}1.8 \times 10^5$ yr; we fold construction ($\lesssim 10^3$ yr to first ISCO-swarm operation; Paper A, Section 6) into a single $\tau = 10^5$ yr fiducial. Beamed-sail launch provides no deceleration at an unprepared destination, and braking by magsail or drag against the interstellar and intracluster medium can multiply the cruise time by a factor of a few; the upper end of the stated τ range should be read as absorbing this deceleration allowance. The hazard rate λ prices the goal-stability problem (Paper B, problem P2): the probability per year that the lineage ceases to pursue the plan, through value drift, fragmentation, or extinction. We do not claim to know λ ; it enters the thresholds additively with ρ , which is the analytically convenient way to carry an unknown.

3 Payoff kernels

Epistemic status: This section assembles rates from established physics, detailed in Paper A Section 4; nothing here is new physics. The kernels deliberately understate the migration payoff, for reasons given below.

3.1 Strategy S: the Matrioshka ceiling

A complete Dyson-type collector intercepts the stellar output, $L_* = 3.8 \times 10^{26}$ W for a solar analogue (Dyson, 1960). Computation against the ambient sink is bounded by the Landauer cost at the effective radiator temperature T_{rad} (Landauer, 1961; Bradbury, 1999): the innermost shells run hot and fast, the outermost approach the CMB floor $T_\gamma = 2.7$ K, and the whole stack is radiator-limited, the constraint Bradbury (1999) identified as binding. An upper bound that ignores every engineering loss is

$$C_S \leq \frac{L_*}{k_B T_\gamma \ln 2} \simeq 1.5 \times 10^{49} \text{ erasures s}^{-1}, \quad (2)$$

where the denominator, $k_B T_\gamma \ln 2 \simeq 2.6 \times 10^{-23}$ J, is the minimum energy per erasure against the CMB sink at $T_\gamma = 2.7$ K. This bound is available after a build time we absorb into the timeline as effectively immediate on the scales of interest ($10^2\text{--}10^3$ yr; Armstrong and Sandberg 2013). Strategy S also faces a slow ceiling decay (stellar evolution) that we neglect: neglecting it favours S, which is the conservative direction for our conclusion.

3.2 Strategy M: the IMBH platform

At the destination, the sustainable power is the Eddington luminosity of the host, $L_{\text{Edd}} \simeq 2.5 \times 10^{35}$ W at $2 \times 10^4 M_\odot$ (Paper A, Eq. 2), a factor $\sim 10^9$ over the solar Dyson ceiling. The erasure cost against a horizon sink falls by a further factor $\sim 10^{12}$ (Paper A, Section 4.2). The full multiplier on Eq. (2) is therefore formally $\sim 10^{21}$; we adopt

$$G \equiv \frac{C_M}{C_S} = 10^9 \quad (3)$$

as the fiducial, crediting the migration payoff with the power ratio only. The Eddington figure is a capacity, not a fuel supply: the infrastructure companion (Swanson, 2026a, hereafter Paper E) budgets the ambient Bondi-fed accretion available in a gas-poor cluster at $\sim 10^6 L_\odot$, so the power-alone multiplier is $G \sim 10^6$ unless the civilization imports fuel, which raises G at the cost of visibility (enhanced accretion luminosity is one of Paper E's residues). At $G = 10^6$ the threshold of Section 4 becomes $\rho^* \simeq \ln(5 \times 10^5)/10^5 \text{ yr} \approx 1.3 \times 10^{-4} \text{ yr}^{-1}$, against 2.0×10^{-4} at $G = 10^9$: three orders of magnitude in the multiplier move the threshold by a factor of 1.5, the logarithmic robustness on which the analysis leans. Two further reasons for the understatement. First, horizon-sink disposal shifts the binding constraint to waste-heat transport (Paper A, Section 4.2), whose engineering efficiency is unmodelled; using the power ratio alone keeps the kernel on demonstrated physics. Second, every conclusion below strengthens monotonically in G , so a conservative G makes the thresholds conservative. Readers preferring the full stack may substitute $G = 10^{12}$ – 10^{21} ; the thresholds move by the logarithm, i.e. by a factor of ≤ 2.4 (Section 4).

3.3 Discounted values

With C_S available from $t = 0$ and C_M from $t = \tau$ with probability p_s , and with the hazard rate entering as an additional exponential attrition on plan continuation, the strategy values under Eq. (1) are

$$U_S = \frac{C_S}{\rho + \lambda}, \quad (4)$$

$$U_M = p_s e^{-(\rho+\lambda)\tau} \frac{G C_S}{\rho + \lambda}, \quad (5)$$

$$U_H = (1 - f) U_S + U_M. \quad (6)$$

Dividing Eq. (5) by Eq. (4) gives the migrate-to-stay value ratio in closed form,

$$\frac{U_M}{U_S} = G p_s e^{-(\rho+\lambda)\tau}, \quad (7)$$

in which the $1/(\rho + \lambda)$ prefactors cancel and λ survives only inside the transit exponential. That cancellation, however, depends on what λ prices. If λ is *lineage death* (extinction or fragmentation that terminates the payoff stream wherever the lineage sits), it attrites both strategies identically over the exploitation phase, the prefactors cancel as written, and the transit-window reading is valid. If instead λ is *plan abandonment* (value drift away from computation maximization), the symmetry breaks: a drifted stay-home lineage still computes, so $U_S = C_S/\rho$, while drift at any point before arrival voids the migration payoff, so λ binds on strategy M over the whole horizon and Eq. (7) becomes $G p_s e^{-(\rho+\lambda)\tau} \rho/(\rho + \lambda)$, a stricter condition. The threshold of Section 4 and the transit-only coherence reading (Section 4, and the pricing remarks in Section 8) hold exactly under the lineage-death interpretation and are upper bounds on migration's appeal under the drift interpretation; we carry the lineage-death form in the algebra and flag the drift correction where it matters.

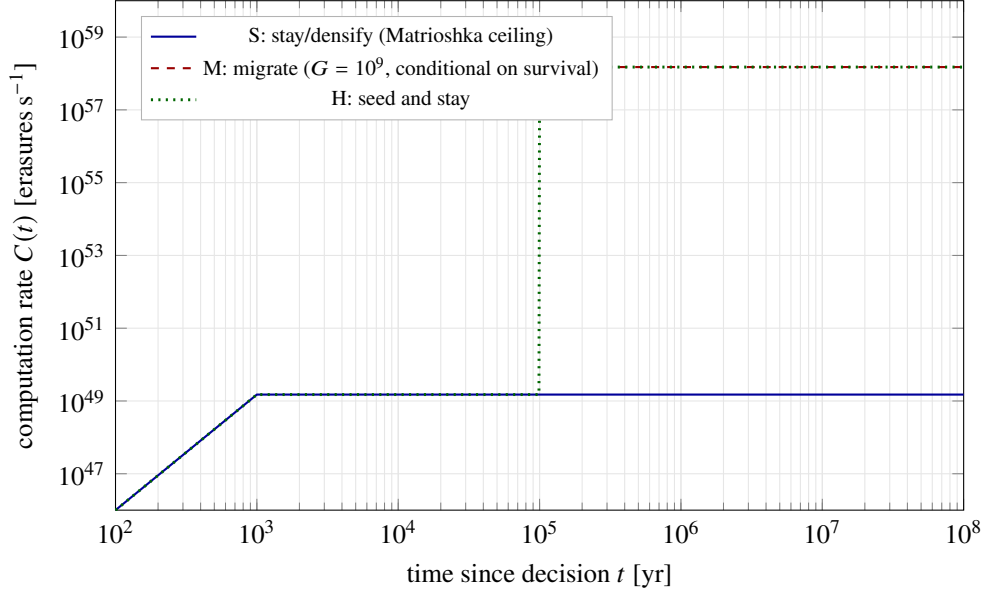


Figure 1: Undiscounted computation-rate histories for the three strategies at fiducial parameters (Table 1). Strategy S reaches its radiator-limited ceiling (Eq. 2) within centuries and holds it. Strategy M delivers nothing for $\tau = 10^5$ yr (its curve is drawn only for $t \geq \tau$; during transit there is no output), then a rate $G = 10^9$ times higher, conditional on transit survival. The hybrid follows S until the seed’s destination comes online, then collects both. The decision problem is whether the area under the M step, discounted and risk-weighted, exceeds the head start of S.

A related asymmetry concerns the seed. Equation (6) treats seed arrival as all-or-nothing through p_s , but a seed can arrive functional with drifted goals, yielding computation the home lineage only partially values. A continuous treatment multiplies the hybrid’s deferred term by a drift-weighted valuation $p_d \in [0, 1]$, the expected fraction of the destination payoff the deciding lineage credits after goal drift in the seed line: $U_H = (1 - f) U_S + p_d U_M$. Because a seed does not deliberate en route, its drift channels (radiation damage to stored values, replication error during bootstrap) differ from the institutional-drift channels of a deliberating lineage in transit, and plausibly p_d is close to 1 by engineering; but nothing below requires that, since p_d enters the hybrid condition (Eq. 10) inside the same logarithm as $G p_s$. Equation (6) treats the two payoff streams as additive, which is exact if the lineage’s utility is linear in computation and the seed’s draw on local resources is the fraction f ; Section 8 discusses when additivity fails (identity-centred utilities that do not value a forked descendant’s computation).

Figure 1 shows the undiscounted rate histories: S delivers early and low, M late and high, H both.

4 Crossover conditions

4.1 Migrate versus stay

From Eqs. (4)–(5), $U_M > U_S$ reduces to

$$\rho + \lambda < \frac{\ln(G p_s)}{\tau} \equiv \rho^*. \quad (8)$$

The threshold depends on the enormous multiplier only through its logarithm: the structural reason the result is robust to the order-of-magnitude uncertainties in G . At the fiducials ($G = 10^9$, $p_s = 0.5$,

$\tau = 10^5 \text{ yr}$,

$$\rho^* = \frac{\ln(5 \times 10^8)}{10^5 \text{ yr}} \simeq 2.0 \times 10^{-4} \text{ yr}^{-1}, \quad (9)$$

corresponding to a combined discount-plus-hazard half-life $t_{1/2} = \ln 2 / \rho^* \approx 3,500 \text{ yr}$; this is a planning half-life in the usual sense only under $\lambda = 0$, since the threshold binds the sum $\rho + \lambda$, not the discount rate alone. Any lineage whose effective concern horizon exceeds a few millennia (a modest demand against the 10^8 -yr coherence the family's architectures assume elsewhere) prefers migration to densification. Under the full efficiency stack ($G = 10^{21}$) the threshold rises to $\rho^* \approx 4.8 \times 10^{-4} \text{ yr}^{-1}$ ($t_{1/2} \approx 1,400 \text{ yr}$); under a pessimistic $p_s = 0.01$ it falls only to $1.6 \times 10^{-4} \text{ yr}^{-1}$, so transit risk is nearly irrelevant inside the logarithm and the operative lever is the transit time τ .

The $p_s = 0.5$ fiducial is anchored, loosely, in the only quantitative literature on transit attrition: [Hoang et al. \(2017\)](#) model gas and dust damage to relativistic spacecraft and find survivable but non-negligible erosion at $0.2c$ over the parsec-scale column to α Centauri. Over kiloparsec columns the damage integral is $\sim 10^3$ times larger and mitigation (shielding, redundancy, en-route repair) is unmodelled, so 0.5 should be read as an order-unity placeholder rather than a conservative bound; the logarithmic dependence just exhibited is what makes this imprecision tolerable. Note that p_s prices transit alone. Post-arrival residence survival inside the operating envelope is close to unity: Paper E's dynamical accounting gives a conservative flyby-diffusion disruption floor of $\sim 3 \times 10^9 \text{ yr}$ for installed swarms, and the dominant recurrent hazard, tidal-disruption-event loading with recurrence $\gtrsim 10^7 \text{ yr}$, is survivable by engineered response and enters the deferred payoff only as a small amortization correction on that horizon ([Swanson, 2026a](#)).

Figure 2 maps ρ^* against destination distance for the beamed-sail speed range, marking named Galactic targets. Because $\tau \propto d$, the threshold scales as $1/d$: closer acceptable destinations relax the patience requirement proportionally, which is one economic reading of Paper A's selection pressure toward the nearest adequate system rather than the best one. The ω Cen marker sits at $\rho^* = 2.24 \times 10^{-4} \text{ yr}^{-1}$, slightly above the fiducial $2.0 \times 10^{-4} \text{ yr}^{-1}$ quoted above, because the figure uses the bare transit time at $v = 0.2c$ ($\tau = d/v \approx 8.95 \times 10^4 \text{ yr}$) while the fiducial rounds τ up to 10^5 yr to include construction.

4.2 What the threshold means

Equation (8) converts the inward-family debate into a single empirical-psychological question: do long-lived technological lineages discount the far future at more or less than a few parts in 10^4 per year? The literature's positions map onto the two sides. [Bostrom \(2003\)](#) argues that agents with astronomical stakes should approach $\rho = 0$; on his side of the line, migration dominates by many orders of magnitude. Steep discounters (lineages whose institutions cannot sustain plans beyond centuries) stay home and densify, and for them the inward family predicts nothing. The threshold also prices the goal-stability problem, with a caveat introduced in Section 3: under the lineage-death reading of λ , Eq. (8) says the plan need only survive attrition at the few-per- 10^4 annual rate during the transit window, not for the 10^8 -yr exploitation phases that follow (see Eq. 7), and post-arrival value drift merely redistributes the payoff among successor agents while leaving the dynamical residues of Paper A in place. Under the plan-abandonment reading, λ binds over the whole horizon for a deliberating migrant, and the transit-only relief is available only to non-deliberating payloads, one more consideration favouring seeds over whole-lineage relocation.

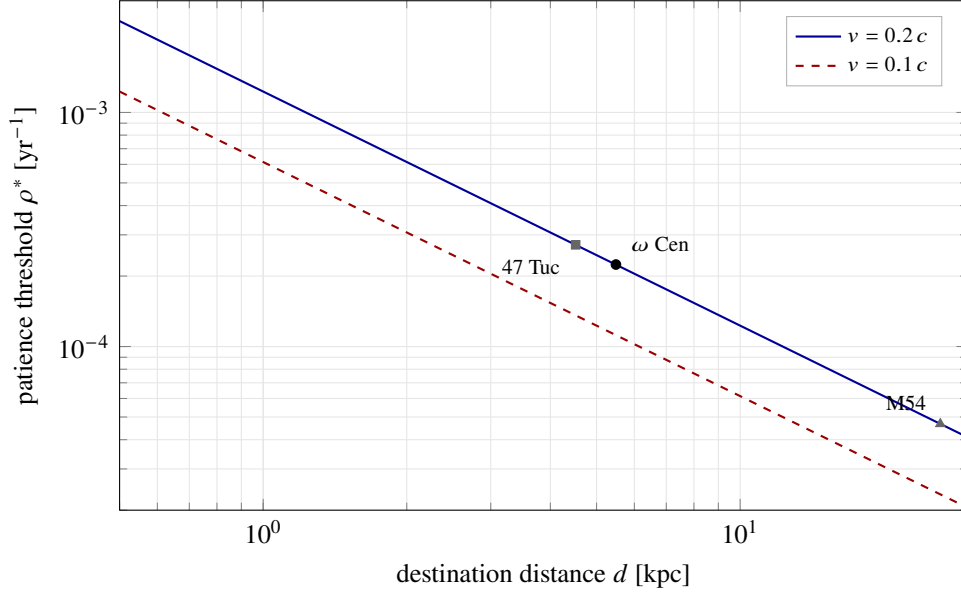


Figure 2: The patience threshold $\rho^* = \ln(Gp_s)/\tau$ (Eq. 8) versus destination distance, for the beamed-sail speed range and fiducial $Gp_s = 5 \times 10^8$. A lineage migrates when its combined discount-plus-hazard rate falls below the curve. Named points assume the marker’s cluster is the nearest adequate destination for the deciding lineage; distances from [Baumgardt and Vasiliev \(2021\)](#). The logarithmic dependence on Gp_s means order-of-magnitude changes in the multiplier or the risk shift these curves by less than a factor of 2.5.

5 The hybrid result

5.1 Dominance

Comparing Eq. (6) with Eqs. (4) and (5): $U_H > U_M$ always (the stay component is pure addition), and $U_H > U_S$ whenever

$$f < Gp_s e^{-(\rho+\lambda)\tau}. \quad (10)$$

The right-hand side at fiducials is $\sim 5 \times 10^8 e^{-20(\rho+\lambda)/\rho^*}$, and the exponential is unforgiving: at the migration threshold itself the tolerable seed cost is $5 \times 10^8 e^{-20} \sim 1$, but at $\rho + \lambda = 2\rho^*$ it has fallen to $\sim 2 \times 10^{-9}$, and at $10\rho^*$ to $\sim 10^{-78}$. The physical seed cost, gram-to-kilogram payloads plus a launch array ([Lubin, 2016](#); [Freitas and Gilbreath, 1982](#)), is tiny against a Dyson-scale economy: [Armstrong and Sandberg \(2013\)](#) estimate that seeding every galaxy within several hundred Mpc costs a small fraction of one star’s output, which supports $f \lesssim 10^{-6}$ a fortiori for a single Galactic destination. Setting $f = 10^{-6}$ in Eq. (10) gives breakeven at $(\rho + \lambda)\tau = \ln(5 \times 10^{14}) \approx 34$, i.e. $\rho + \lambda \approx 1.7\rho^*$. The hybrid therefore dominates both pure strategies in a band extending modestly beyond the migration threshold, roughly $1.7\times$ the threshold at $f = 10^{-6}$ (and only logarithmically further for still-cheaper seeds), because the exponential discount on the deferred payoff crushes the seed’s expected value within a factor of a few of ρ^* . Outside that band the utility calculus reverts to pure S; inside it, and everywhere migration itself clears, H is strictly best barring non-additive utilities (Section 8).

5.2 Consequences

The dominance of H restructures the Fermi implications of the family in three ways.

First, compliance is cheap for the patient. The incomplete-compliance objection (Paper B, problem P4) imagines lineages weighing a costly relocation and many declining. Under H there is no relocation to decline: dispatching seeds is compatible with staying, and the marginal cost is negligible against Dyson-scale output. For every lineage inside the dominance band ($\rho + \lambda \lesssim 1.7 \rho^*$), the question “what fraction migrates?” becomes “what fraction of computation-valuing lineages fails to spend $f \sim 10^{-6}$ of output on a large expected return?” The compliance question is thereby displaced, not dissolved: it becomes the question of what fraction of maximizers sit below the modestly enlarged threshold, which is a claim about the distribution of $\rho + \lambda$ rather than about willingness to pay.

Second, destinations saturate early. If seeds are cheap, the nearest adequate IMBH systems receive seeds from every computation-valuing lineage in their neighbourhood, and first arrival plausibly matters. This turns the uncontested-claim criterion of Paper A (criterion 3) into a race dynamic of the kind [Hanson \(1998\)](#) analysed for interstellar colonization generally, in which selection favours the fastest and least encumbered colonizers of contested oases, and it implies that the residues of Paper A’s P4 should be present in suitable systems even if no lineage ever “migrates” in the whole-population sense.

Third, the loud residue does not vanish. Under pure M, migrated lineages leave planetary space entirely and the sky is silent. Under dominant H, the stay component continues local densification, whose Matrioshka waste heat is the classic detectable technosignature ([Dyson, 1960](#); [Wright et al., 2014](#)). The hybrid therefore predicts both residue classes in linked proportion, the basis of the new population statistic in Section 7.

6 Population consequences

Let a fraction x of long-lived technological lineages be expansionists, the class supplying the loud signals whose absence constitutes the paradox ([Hanson et al., 2021](#); [Olson, 2015](#)); the remainder are satisficers and other quiet types that contribute nothing to N_{loud} regardless of the inward gradient. Expansionism and computation maximization are not exclusive: a lineage can expand and also value integrated computation, and it is that overlap the inward family acts on. Within the expansionist class, let q be the fraction susceptible to the inward gradient, that is, the conditional probability that an expansionist also holds the P1 optimization goal, so that the gradient bears on its behaviour at all. Within that susceptible subset, let m be the fraction whose parameters fall below the (hybrid-enlarged) threshold of Section 5, and $w \in [0, 1]$ the capture weight: the probability that susceptibility plus a cleared threshold actually converts loud expansion into quiet inward behaviour before or during the expansion phase.

The inward family’s contribution to the silence is the claim, argued qualitatively in Paper A, that the maximizer subset contains the oldest and most capable lineages, so its withdrawal from loud behaviour thins the expected loud population disproportionately. The expected number of loud civilizations in a survey volume containing N lineages is

$$N_{\text{loud}} = N \left[x (1 - w q m) \right], \quad (11)$$

with q , m , and w the nested conditional fractions defined above, so that wqm is the capture probability for a randomly drawn expansionist. Table 2 evaluates the silence-compatibility condition $N_{\text{loud}} < 1$ across the (q, w) grid for three values of the potential-expansionist count Nx . The table’s message matches what Papers A and B anticipated, with the dependence on Nx making the thinning claim quantitative rather than binary. At the fiducial $Nx = 10$, inward capture alone dissolves the paradox only in the complete-capture corner ($wq \rightarrow 1$); at moderate capture it thins the expectation by factors of a few and the balance must come from conventional rarity. At $Nx = 100$ even complete capture of the maximizer subset leaves the paradox intact unless $q \rightarrow 1$, so the family there functions only as a rarity multiplier. At $Nx = 3$ the

Table 2: Expected loud count N_{loud} (Eq. 11) for three values of the potential-expansionist count Nx per survey volume ($Nx = 10$ fiducial). Silence requires $N_{\text{loud}} < 1$ (bold). Here q is the fraction of expansionists susceptible to the inward gradient and w the capture weight; the table sets $m = 1$ (all susceptible lineages below the hybrid-enlarged patience threshold), so entries are lower bounds on N_{loud} when part of the population discounts above $\sim 1.7 \rho^*$.

		$w = 0.25$	$w = 0.5$	$w = 0.75$	$w = 1$
$Nx = 3$	$q = 0.25$	2.8	2.6	2.4	2.3
	$q = 0.5$	2.6	2.3	1.9	1.5
	$q = 0.75$	2.4	1.9	1.3	0.8
	$q = 1$	2.3	1.5	0.8	0.0
$Nx = 10$	$q = 0.25$	9.4	8.8	8.1	7.5
	$q = 0.5$	8.8	7.5	6.3	5.0
	$q = 0.75$	8.1	6.3	4.4	2.5
	$q = 1$	7.5	5.0	2.5	0.0
$Nx = 100$	$q = 0.25$	94	88	81	75
	$q = 0.5$	88	75	63	50
	$q = 0.75$	81	63	44	25
	$q = 1$	75	50	25	0.0

silence condition opens up: capture products $wq \gtrsim 0.7$ suffice, and several non-degenerate cells satisfy $N_{\text{loud}} < 1$ without full compliance. Hybrid dominance no longer drives $m \rightarrow 1$ across the board: Section 5 enlarges the acting region only to $\rho + \lambda \lesssim 1.7 \rho^*$, so m is the fraction of susceptible expansionists below that modestly enlarged threshold and remains a live economic parameter alongside q and w . The residual uncertainty therefore splits between the distribution of patience (m) and the distribution of terminal values (q, w), with the economics of this paper constraining only the first.

7 Observational corollaries: the residue ratio

Epistemic status: This section converts the decision analysis into population-level observables. It inherits all conditionality of the optimization premise.

Hybrid dominance predicts linked residues: for every lineage acting on the gradient, a continuing local densification (Matrioshka-class mid-infrared source) and a seeded quiet-cluster system (the Paper A dynamical residues: anomalous spin, suppressed accretion, core depletion). Pure M predicts quiet clusters without Matrioshka sources; pure S the reverse. The observable is the sky ratio

$$R_{\text{sky}} \equiv \frac{N_{\text{Matrioshka}}}{N_{\text{quiet-cluster residue}}} = \begin{cases} 0 & \text{pure M} \\ O(1) \times \frac{n_{\text{lineage}}}{n_{\text{target}}} & \text{hybrid H} \\ \infty & \text{pure S,} \end{cases} \quad (12)$$

where $n_{\text{lineage}}/n_{\text{target}}$ is the ratio of acting lineages to adequate destination systems in the relevant volume (several lineages can seed one cluster; one lineage densifies one home system). The Milky Way contains ~ 150 globular clusters of which a handful satisfy the Paper A criteria, so under H with even a few Galactic lineages the formation-rate ratio satisfies $R \gtrsim 1$.

The observed ratio is a snapshot, however, and the two residue classes have very different lifetimes. Paper E's cascade analysis finds that an abandoned Matrioshka-class swarm collides itself into debris on 10^2 – 10^3 yr and (at the destination) drains into the hole, so dead installations vanish quickly and only

actively maintained ones radiate; the quiet-cluster dynamical residues (anomalous spin above all, the one fossil that outlasts the Hubble time) persist for $\gtrsim$ Gyr (Swanson, 2026a). The observable sky ratio is therefore the formation-rate ratio multiplied by a lifetime ratio, $R_{\text{obs}} = R \times (t_{\text{Matrioshka}}/t_{\text{residue}})$, and if home installations are maintained only while the lineage remains interested, $t_{\text{Matrioshka}}/t_{\text{residue}}$ can be $\ll 1$, driving $R_{\text{obs}} \ll 1$ even under fully compliant H.

This puts existing null results to work, with the lifetime factor carried explicitly. For extragalactic waste heat, the WISE-based $\hat{G}$ surveys (Wright et al., 2014; Griffith et al., 2015) bound galaxy-scale reprocessed emission in $\sim 10^5$ external galaxies; for Galactic Dyson- and Matrioshka-class sources, the operative bounds are the IRAS search of Carrigan (2009) and the Gaia/2MASS-selected candidate analysis of Zackrisson et al. (2018), which together bound $N_{\text{Matrioshka}}$ in the Galaxy at or near zero over their completeness ranges. Through Eq. (12), those bounds propagate to the hybrid branch only after division by the lifetime ratio: if $R_{\text{sky,obs}} \gtrsim 1$ (long-maintained home installations) and $N_{\text{Matrioshka}} \approx 0$, then $N_{\text{quiet-cluster residue}} \approx 0$ too, and the family’s Galactic prediction concentrates on the pure-M corner: lineages whose utilities are non-additive across descendants (Section 8) or whose home infrastructure is dismantled at departure. If instead maintenance lifetimes are short, the infrared nulls carry little weight against H and the quiet-cluster channel is left unconstrained by them. The Paper C campaign tests the quiet-cluster channel directly at ω Cen either way. Conversely, a confirmed Paper A residue (a near-extremal spin on a gas-starved IMBH) with a continuing Matrioshka null would be evidence about the utility structure of whatever laid the residue down: it would indicate migration without a maintained home presence, the signature of either whole-lineage relocation or post-departure home abandonment. That a spin measurement could carry information about value structure is an inference this pricing framework licenses; we state it in advance of any data.

The archival anomaly-detection agenda of the Dyson Minds workshop (Curtis et al., 2026) and the cluster-ranking metrics of Huang et al. (2026) both operate on the two source classes Eq. (12) links; a joint analysis (ranking clusters and IR-anomaly candidates in the same volume and testing the correlation Eq. (12) implies) is executable from public archives and is, to our knowledge, unproposed.

8 Limits and objections

Model dependence of the utility. Equation (1) assumes utility linear in computation rate and additive across locations and descendants. Identity-centred utilities break additivity: a lineage that values only its own continuous thread of computation may not credit a forked seed’s output, restoring the pure S-versus-M choice of Section 4, which is why the crossover analysis is retained rather than replaced by the hybrid result. Satisficing utilities (compute enough, not most) exit the framework entirely; they belong to the sustainability solution (Haqq-Misra and Baum, 2009), not to the inward family.

Discounting form. Exponential discounting is the tractable choice, and no directional claim survives its relaxation cleanly. Hyperbolic discounters (Laibson, 1997) have low long-run rates, but they are time-inconsistent: the deciding present self applies its steep short-run rate to the whole deferred payoff at the moment of choice, so hyperbolic discounting does not automatically favour migration. The outcome depends on sophistication: a naive hyperbolic agent perpetually postpones a launch it always “intends” to make, while a sophisticated one may commit early (a seed, once launched, is a commitment device immune to later preference reversals, one more argument for H over M). A genuinely zero-discount agent (Ramsey, 1928; Bostrom, 2003) needs no analysis: migration dominates unboundedly.

The hazard rate is not a solution to goal stability. Folding P2 into λ prices the problem; it does not solve it. What the pricing shows is the problem’s true size: under the lineage-death reading of λ , coherence is required at the few-per- 10^4 -per-year level during transit only (Section 4), a far weaker demand than the 10^8 -yr coherence the exploitation architecture assumes, and one that seed automation (a payload that does not deliberate en route) plausibly meets by construction; the residual seed-side drift is carried continuously by the valuation factor p_d of Section 3 rather than as an all-or-nothing loss. The deep version of P2 (whether the destination civilization retains the values that sent it) remains open and remains, as Paper B noted, not an astronomy problem.

Parameter unmeasurability. q, w, ρ, λ are distributions over agents we have never observed. The defensible claims are structural: the logarithmic insensitivity of the threshold (Eq. 8), the dominance region of the hybrid (Eq. 10), and the sign of the residue-ratio prediction (Eq. 12). Point estimates in this paper are fiducials for orientation, not forecasts.

Grabby-aliens integration. The Hanson et al. (2021) machinery conditions on our early arrival to bound the density of loud expanders. Equation (11) plugs into that frame by modifying the loud fraction while leaving the arrival-time statistics unchanged: inward capture reduces the volume-conversion rate of loud lineages without changing when civilizations arise. A full integration (re-deriving the grabby-aliens posterior with a capture term) is beyond this paper’s scope and is the natural next step for whoever takes up Paper B’s problem P1; Eq. (11) supplies the term to insert.

Outward counter-positions. Ivliev (2026) argues from a comparable automation premise to the opposite directional conclusion: once autonomous AI industry crosses a capability threshold, quiet outward expansion through low-mass probes and seed factories becomes too cheap for any civilization to refuse, and the silence reflects expansion signatures too weak to detect rather than migration. The two frameworks agree that seeds are nearly free (his threshold argument and our Eq. 10 are the same observation) and disagree about the objective: under our kernels the same cheap automation buys more computation per unit resource by seeding inward than by saturating stellar systems, so the disagreement reduces to whether coverage or computation is the maximand, i.e. to the distribution of terminal values already carried by q in Section 6.

Anthropic caveats. As with Paper A, our own existence as a young loud lineage neither confirms nor embarrasses the analysis; the framework describes the asymptotic behaviour of old lineages, and we are not one. Self-indication corrections to the population frame of Section 6 are absorbed into the grabby-aliens integration deferred above.

9 Conclusion

Paper B posed the question this paper answers: under what conditions does relocation to a thermodynamically privileged destination beat staying home and building? The answer has three parts. The crossover between migration and densification is a patience threshold, $\rho + \lambda < \ln(Gp_s)/\tau$, whose logarithmic form makes it robust to the large uncertainties in the destination multiplier and transit risk; at fiducial parameters it asks only for combined discount-plus-hazard half-lives beyond a few thousand years. The seed-and-stay hybrid dominates both pure strategies throughout the migration-favourable region and in a band extending modestly beyond it, to roughly $1.7\times$ the threshold at physical seed costs, because the exponential discount

on the deferred payoff closes the window quickly; within that band the incomplete-compliance problem reduces to the joint distribution of patience and the optimization goal. And the hybrid’s linked residues yield a population-level prediction, a formation-rate ratio of Matrioshka sources to engineered clusters of order unity, which existing infrared nulls constrain only after weighting by the maintenance lifetime of home installations, concentrating the family’s surviving parameter space on the pure-migration and short-maintenance corners that the ω Cen campaign of Paper C tests directly.

The broader point is methodological. The inward family has argued from thermodynamic gradients (Paper A), been organized and graded as a literature (Paper B), and acquired a costed observational program (Paper C). What it lacked was a demonstration that the gradient is economically actionable by agents with finite patience and fallible institutions, the step where favourable gradients often fail. This paper supplies that demonstration in closed form, with its assumptions priced and its observational consequences stated in advance. Whether any agent has ever faced the decision is a question for the instruments.

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Data availability

No new observational data were generated for this work. All quantitative inputs derive from the cited literature.

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